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Doulos does not endorse training materials from other suppliers on EDA Playground.

▼ Languages & Libraries

Testbench + Design

SystemVerilog/Verilog

UVM / OVM

None

Other Libraries

0 selected

☐ Enable TL-Verilog☐ Enable Easier UVM☐ Enable VUnit

▼ Tools & Simulators

Icarus Verilog 12.0

Compile Options

-Wall -g2012

Run Options

Run Options

☐ Use run.bash shell script☒ Open EPWave after run☐ Show output text file☐ Show output SVG file☐ Download files after run

▼ Examples

using EDA Playground

VHDL

Verilog/SystemVerilog

UVM

EasierUVM

SVAUnit

SVUnit

VUnit (Verilog/SV)

VUnit (VHDL)

TL-Verilog

e + Verilog

Python + Verilog Python +

VHDL

Python Only

C++/SystemC

236

testbench.sv

```

1 module and_gate_tb;
2 reg a, b;
3 wire y;
4
5 and_gate uut(.a(a), .b(b), .y(y));
6
7 initial begin
8     $dumpfile("dump.vcd");
9     $dumpvars(0, and_gate_tb);
10 end
11
12 initial begin
13     a=0; b=0;
14     #10 a=0; b=1;
15     #10 a=1; b=0;
16     #10 a=1; b=1;
17     #10;
18     $finish;
19 end
20
21 endmodule

```

design.sv

```

1 // Code your design here
2 module and_gate(input a, input b, output
3 y);
4 assign y = a & b;
5 endmodule

```

Log

Share

[2025-06-30 09:23:30 UTC] iverilog -Wall -g2012 design.sv testbench.sv && unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

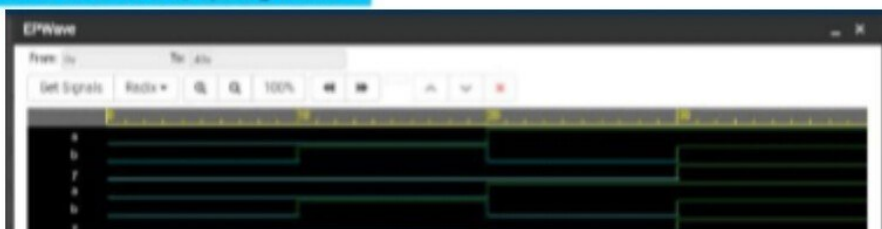
testbench.sv:20: \$finish called at 40 (1s)

Finding VCD file...

./dump.vcd

[2025-06-30 09:23:31 UTC] Opening EPWave

Done



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Languages & Libraries

Testbench + Design

SystemVerilog/Verilog

UVM / OVM

None

Other Libraries

0 selected

- ☐ Enable TL-Verilog
- ☐ Enable Easier UVM
- ☐ Enable VUnit

Tools & Simulators

Icarus Verilog 12.0

Compile Options

Wall-g2012

Run Options

Run Options

- ☐ Use run.bash shell script
- ☒ Open EPWave after run
- ☐ Show output text file
- ☐ Show output SVG file
- ☐ Download files after run

Examples

- using EDA Playground
- VHDL
- Verilog/SystemVerilog
- UVM
- EasierUVM
- SVAUnit
- SVUnit
- VUnit (Verilog/SV)
- VUnit (VHDL)
- TL-Verilog
- e + Verilog
- Python + Verilog Python + VHDL
- Python Only
- C++/SystemC

236

testbench.sv

```

1 // Code your testbench here
2 // or browse Examples
3 module or_gate_tb;
4 reg a, b;
5 wire y;
6
7 or_gate uut(.a(a), .b(b), .y(y));
8
9 initial begin
10     $dumpfile("dump.vcd");
11     $dumpvars(0, or_gate_tb);
12 end
13
14 initial begin
15     a=0; b=0;
16     #10 a=0; b=1;
17     #10 a=1; b=0;
18     #10 a=1; b=1;
19     #10;
20     $finish;
21 end
22
23 endmodule

```

design.sv

```

1 // Code your design here
2 module or_gate(input a, input b, output
3 y);
4 assign y = a | b;
5 endmodule

```

Log Share

[2025-06-30 09:15:38 UTC] iverilog "-Wall" "-g2012" design.sv testbench.sv -BA unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

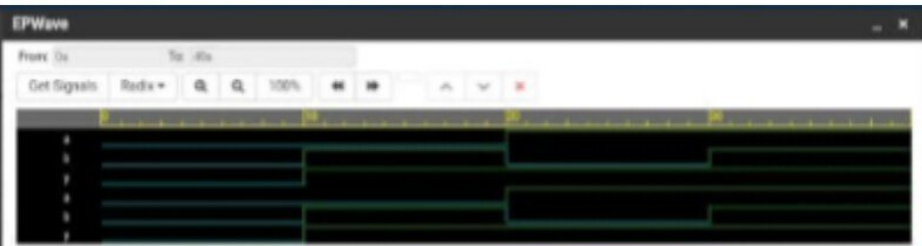
testbench.sv:20: \$finish called at 40 (1s)

Finding VCD file...

./dump.vcd

[2025-06-30 09:15:39 UTC] Opening EPWave...

Done



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▼ Languages & Libraries

Testbench + Design

SystemVerilog/Verilog ▾

UVM / OVM ⓘ

None ▾

Other Libraries ⓘ

0 selected ▾

- ☐ Enable TL-Verilog ⓘ
- ☐ Enable Easier UVM ⓘ
- ☐ Enable VUnit ⓘ

▼ Tools & Simulators ⓘ

Icarus Verilog 12.0 ▾

Compile Options

-Wall -g2012

Run Options

Run Options

- ☐ Use `run.bash` shell script
- ☒ Open EPWave after run
- ☐ Show output text file
- ☐ Show output SVG file
- ☐ Download files after run

▼ Examples

using EDA Playground

VHDL

Verilog/SystemVerilog

UVM

EasierUVM

SVAUnit

SVUnit

VUnit (Verilog/SV)

VUnit (VHDL)

TL-Verilog

e + Verilog

Python + Verilog Python + VHDL

Python Only

C++/SystemC

236

testbench.sv

```
1 module not_gate_tb;
2 reg a;
3 wire y;
4 not_gate uut(.a(a),.y(y));
5
6 initial begin
7     $dumpfile("dump.vcd");
8     $dumpvars(0,not_gate_tb);
9 end
10
11 initial begin
12     a=0;
13     #10 a=1;
14     #10 $finish;
15 end
16 endmodule
```

design.sv

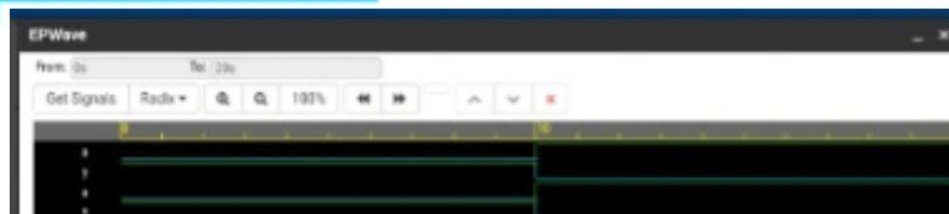
```
1 // Code your design here
2 module not_gate(input a, output y);
3     assign y = ~a;
4 endmodule
```

Log Share

```
[2025-06-30 09:39:21 UTC] iverilog -Wall -g2012 design.sv testbench.sv && unbuffer vvp a.out
VCD info: dumpfile dump.vcd opened for output.
testbench.sv:16: $finish called at 20 (1s)
Finding VCD file...
./dump.vcd
```

[2025-06-30 09:39:21 UTC] Opening EPWave

Done



to you by **DOULOS**your online working environment from
an EDA Playground

Languages & Libraries

Language + Design

Verilog/Verilog

DVM

Libraries

ed

le TL-Verilog

le Easier UVM

le VUnit

Tools & Simulators

Iverilog 12.0

Simulation Options

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Simulation Options

Simulation

un.bash shell script

EPWave after run

| output text file

| output SVG file

| load files after run

Examples

EDA Playground

ig/SystemVerilog

rUVM

init

it

(Verilog/SV)

(VHDL)

inlog

erilog

in + Verilog Python +

in Only

SystemC

testbench.sv

```
1 module nand_gate_tb;
2 reg a, b;
3 wire y;
4
5 nand_gate uut(.a(a), .b(b), .y(y));
6
7 initial begin
8     $dumpfile("dump.vcd");
9     $dumpvars(0, nand_gate_tb);
10 end
11
12 initial begin
13     a=0; b=0;
14     #10 a=0; b=1;
15     #10 a=1; b=0;
16     #10 a=1; b=1;
17     #10 $finish;
18 end
19 endmodule
```

design.sv

```
1 // Code your design here
2 module nand_gate(input a, input b, output
3 y);
4 assign y = ~(a & b);
5 endmodule
```

Log

Share

[2026-06-30 09:32:29 UTC] iverilog -Wall -g2012 design.sv testbench.sv -s unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

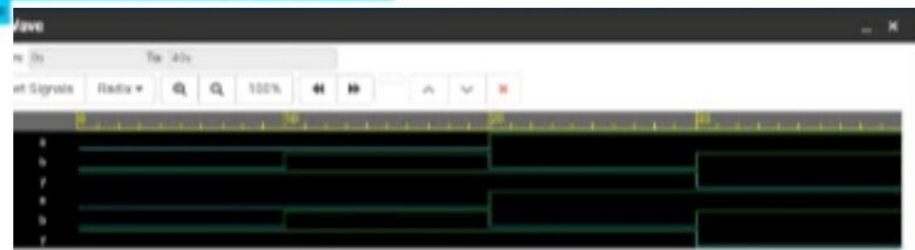
testbench.sv:19: \$finish called at 40 (1s)

Finding VCD file...

./dump.vcd

[2026-06-30 09:32:29 UTC] Opening EPWave...

Done



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Doulos test and design training market at Fraunhofer IPA EDA Playground

Languages & Libraries

bench + Design

imVerilog/Verilog

I/OVM

r Libraries

ected

able TL-Verilog

able Easier UVM

able VUnit

Tools & Simulators

s Verilog 12.0

File Options

g2012

Options

Options

e run.bash shell script

ien EPWave after run

ow output text file

ow output SVG file

wload files after run

Examples

ng EDA Playground

IDL

rilog/SystemVerilog

M

sierUVM

AUnit

Unit

nit (Verilog/SV)

nit (VHDL)

Verilog

Verilog

thon + Verilog Python +

IDL

thon Only

+/SystemC

testbench.sv

```
1 // Code your testbench here
2 // or browse Examples
3 module nor_gate_tb;
4 reg a, b;
5 wire y;
6
7 nor_gate uut(.a(a), .b(b), .y(y));
8
9 initial begin
10     $dumpfile("dump.vcd");
11     $dumpvars(0, nor_gate_tb);
12 end
13
14 initial begin
15     a=0; b=0;
16     #10 a=0; b=1;
17     #10 a=1; b=0;
18     #10 a=1; b=1;
19     #10 $finish;
20 end
21
22 endmodule
```

design.sv

```
1 // Code your design here
2 module nor_gate(input a, input b, output
3 y);
4 assign y = ~(a | b);
5 endmodule
```

Log Share

[2026-06-30 09:37:01 UTC] iverilog '-Wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

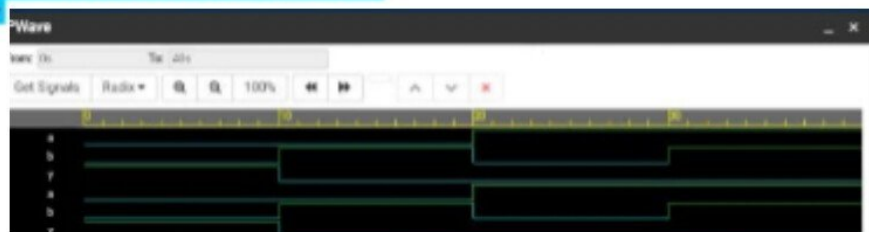
testbench.sv:19: \$finish called at 40 (1s)

Finding VCD file...

./dump.vcd

[2026-06-30 09:37:02 UTC] Opening EPWave...

Done



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▼ Languages & Libraries

Testbench + Design

SystemVerilog/Verilog ▾

UVM / OVM ⓘ

None ▾

Other Libraries ⓘ

0 selected ▾

☐ Enable TL-Verilog ⓘ☐ Enable Easier UVM ⓘ☐ Enable VUnit ⓘ

▼ Tools & Simulators ⓘ

Icarus Verilog 12.0 ▾

Compile Options

-Wall -g2012

Run Options

Run Options

☐ Use `run.bash` shell script☒ Open EPWave after run☐ Show output text file☐ Show output SVG file☐ Download files after run

▼ Examples

using EDA Playground

VHDL

Verilog/SystemVerilog

UVM

EasierUVM

SVAUnit

SVUnit

VUnit (Verilog/SV)

VUnit (VHDL)

TL-Verilog

e + Verilog

Python + Verilog Python +

VHDL

Python Only

C++/SystemC

236

testbench.sv

```
1 module xor_gate_tb;
2 reg a, b;
3 wire y;
4
5 xor_gate uut(.a(a), .b(b), .y(y));
6
7 initial begin
8     $dumpfile("dump.vcd");
9     $dumpvars(0, xor_gate_tb);
10 end
11
12 initial begin
13     a=0; b=0;
14     #10 a=0; b=1;
15     #10 a=1; b=0;
16     #10 a=1; b=1;
17     #10 $finish;
18 end
19
20 endmodule
```

design.sv

```
1 // Code your design here
2 module xor_gate(input a, input b, output
3 y);
4 assign y = a ^ b;
5 endmodule
```

Log

Share

[2026-06-30 09:42:41 UTC] iverilog "-Wall" "-g2012" design.sv testbench.sv -s unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

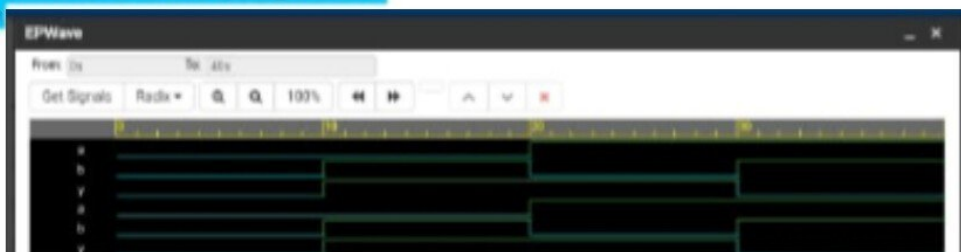
testbench.sv:19: \$finish called at 40 (1s)

Finding VCD file...

./dump.vcd

[2026-06-30 09:42:42 UTC] Opening EPWave...

Done



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▼ Languages & Libraries

Testbench + Design

SystemVerilog/Verilog ▼

UVM / OVM ⓘ

None ▼

Other Libraries ⓘ

0 selected ▼

☐ Enable TL-Verilog ⓘ

☐ Enable Easier UVM ⓘ

☐ Enable VUnit ⓘ

▼ Tools & Simulators ⓘ

Icarus Verilog 12.0 ▼

Compile Options

-Wall -g2012

Run Options

Run Options

☐ Use `run.bash` shell script

☒ Open EPWave after run

☐ Show output text file

☐ Show output SVG file

☐ Download files after run

▼ Examples

using EDA Playground

VHDL

Verilog/SystemVerilog

UVM

EasierUVM

SVAUnit

SVUnit

VUnit (Verilog/SV)

VUnit (VHDL)

TL-Verilog

e + Verilog

Python + Verilog Python +

VHDL

Python Only

C++/SystemC

236

testbench.sv

```
1 // Code your testbench here
2 // or browse Examples
3 module half_adder_tb;
4 reg a, b;
5 wire sum, carry;
6
7 half_adder uut(.a(a), .b(b), .sum(sum),
8               .carry(carry));
9
10 initial begin
11     $dumpfile("dump.vcd");
12     $dumpvars(0, half_adder_tb);
13 end
14
15 initial begin
16     a=0; b=0;
17     #10 a=0; b=1;
18     #10 a=1; b=0;
19     #10 a=1; b=1;
20     #10 $finish;
21 end
22 endmodule
```

design.sv

```
1 // Code your design here
2 module half_adder(input a, input b,
3 output sum, output carry);
4 assign sum = a ^ b;
5 assign carry = a & b;
6 endmodule
```

Log Share

[2026-06-30 09:45:22 UTC] iverilog -Wall -g2012 design.sv testbench.sv && unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

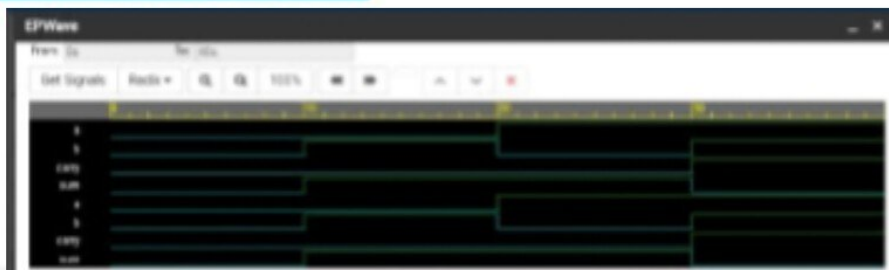
testbench.sv:19: \$finish called at 40 (1s)

Finding VCD file...

./dump.vcd

[2026-06-30 09:45:23 UTC] Opening EPWave...

Done



Languages & Libraries

Testbench + Design

SystemVerilog/Verilog

UVM / OVM

None

Other Libraries

0 selected

- ☐ Enable TL-Verilog
- ☐ Enable Easier UVM
- ☐ Enable VUnit

Tools & Simulators

Icarus Verilog 12.0

Compile Options

-Wall -g2012

Run Options

Run Options

- ☐ Use run.bash shell script
- ☒ Open EPWave after run
- ☐ Show output text file
- ☐ Show output SVG file
- ☐ Download files after run

Examples

- using EDA Playground
- VHDL
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- UVM
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- SVAUnit
- SVUnit
- VUnit (Verilog/SV)
- VUnit (VHDL)
- TL-Verilog
- e + Verilog
- Python + Verilog Python + VHDL
- Python Only
- C++/SystemC

236

testbench.sv

```

1 // Code your testbench here
2 // or browse Examples
3 module full_adder_tb;
4 reg a, b, cin;
5 wire sum, carry;
6
7 full_adder uut(.a(a), .b(b), .cin(cin),
8               .sum(sum), .carry(carry));
9
10 initial begin
11     $dumpfile("dump.vcd");
12     $dumpvars(0, full_adder_tb);
13 end
14
15 initial begin
16     a=0; b=0; cin=0;
17     #10 a=0; b=0; cin=1;
18     #10 a=0; b=1; cin=0;
19     #10 a=0; b=1; cin=1;
20     #10 a=1; b=0; cin=0;
21     #10 a=1; b=0; cin=1;
22     #10 a=1; b=1; cin=0;
23     #10 a=1; b=1; cin=1;
24     #10 $finish;
25 end
26
27 endmodule

```

design.sv

```

1 // Code your design here
2 module full_adder(input a, input b, input
3 cin,
4               output sum, output
5 carry);
6
7 assign sum = a ^ b ^ cin;
8 assign carry = (a & b) | (b & cin) | (a &
9 cin);
10
11 endmodule

```

Log <Share

[2026-06-30 09:49:34 UTC] iverilog -Wall -g2012 design.sv testbench.sv -A& unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

testbench.sv:24: \$finish called at 80 (1s)

Finding VCD file...

./dump.vcd

[2026-06-30 09:49:35 UTC] Opening EPWave

Done

